Muneeb Pervez

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Education

Lahore University of Management Sciences (LUMS) **2023 – 2027**
Bachelor of Science in Computer Science *Lahore, Pakistan*

- Dean's Honor List: Academic Years 2023–24, 2024–25, 2025–26, and 2026–27

Accepted Publications

[1] **Muhammad Muneeb Pervez**, Muhammad Qasim Atiq Ullah, Ibrahim Ahmed Khan, Roshnik Rahat, Muhammad Fareed Zaffar, Rashid Tahir, Talal Rahwan, Yasir Zaki. *(Mis-)Informed Consent: Predatory Apps and the Exploitation of Populations with Limited Literacy.* **The Web Conference 2026.** DOI: 10.1145/3774904.3792969.

[2] Wajiha Naveed, **Muhammad Muneeb Pervez***, Zaeem Mohtashim Khan*, Zafar Ayyub Qazi, Zartash Afzal Uzmi. *Separating Clicks from Baits: Using Large Language Models to Detect Misleading YouTube Thumbnails.* **AAAI ICWSM 2027.** DOI: 10.48550/arXiv.2607.23739.

Experience

Research Assistant - Internet Security and Privacy Lab (ISPL) **June 2025 – Present**
Lahore University of Management Sciences (LUMS) *Supervisor: Dr Fareed Zaffar*

[1] Co-authored a peer-reviewed study of 50 predatory loan, gambling, and trading applications, combining static application analysis, privacy-disclosure auditing, and a targeted user study to examine the exploitation of users with limited literacy. Evaluated LLM-generated summaries, translations, and visual cues as scalable privacy-literacy interventions, finding that 85% of participants did not understand basic application permissions. Accepted at **The Web Conference 2026**.

Research Assistant - Distributed and AI Systems (DAIS) Lab **March 2026 – Present**
Lahore University of Management Sciences (LUMS) *Supervisor: Dr Zafar Ayyub Qazi*

- Co-developed **safeMAS**, a measurement framework for testing whether splitting tools across a team of LLM agents limits the damage of attacks, with 1,110 scenarios in four simulated domains, run on three models. Found that orchestrated teams are attacked more often than a single agent (62% against 48%) and that a tampered message from the orchestrator succeeds in 82% of runs. Under review at **ICLR 2027.**
- Currently extending **safeMAS** to **internal threats**: cases where an agent, given a legitimate task and no external attacker, reasons its way into harmful, unsanctioned actions. Studying whether multi-agent architectures and tool partitioning change how often this happens and how much damage it causes.

[2] Co-authored a cross-cultural study of 2,843 YouTube videos from eight countries, developing a multimodal LLM pipeline that analyzes thumbnails, video descriptions, and subtitle transcripts to detect misleading thumbnails. Benchmarked four frontier LLMs and two open-weight vision-language models, with the strongest configuration achieving 93.8% accuracy. Accepted at **AAAI ICWSM 2027.**

- Measuring how Google's AI Overviews divide the results page between advertisers and publishers, across 10,068 queries on desktop and mobile. Early results show ads inside about one in ten AI Overviews. Planned for **IMC 2027**.

Teaching Assistant - CS5803: Applied Cryptography **Fall 2026**

Teaching Assistant - CS225: Fundamentals of Computer Systems **Fall 2026**

Teaching Assistant - CS100: Computational Problem Solving **Summer 2026**

Head Teaching Assistant - CS100: Computational Problem Solving **Spring 2026**

- Led a team of 12 TAs, overseeing recruitment, scheduling, and grading standards for a student cohort of 250.

Teaching Assistant - CS382: Network-Centric Computing **Spring 2026**

Teaching Assistant - CS100: Computational Problem Solving **Summer 2025**

Relevant Coursework

- Topics in Network Security
- Applied Cryptography
- Compiler Design
- Network Security
- Distributed Systems
- Network-Centric Computing
- Operating Systems
- Big Data Analytics
- AI and Machine Learning

Research Interests

LLM and Multi-Agent Security, Measurements, Systems Security

Projects

Software Vulnerability Detection & Exploitation | *C, x86 Assembly, GDB, Python* **Fall 2025**

- Performed hands-on static and dynamic analysis to identify memory safety vulnerabilities: constructed stack layouts, developed shellcode, and applied NOP sled techniques to achieve reliable exploitation of stack-based buffer overflows.
- Analyzed compiled binaries using GDB to map memory regions, identify control flow hijack points, and validate exploit reliability. Experience directly relevant to automated vulnerability detection and program analysis research.

Web Application Security Analysis | *Python, BurpSuite, FastAPI* **Spring 2026**

- Identified and exploited CSRF and stored/reflected XSS vulnerabilities through systematic input validation testing and HTTP request interception; subsequently implemented defenses including CSRF tokens, input sanitization, and CSP headers.
- Bypassed authentication and access control mechanisms through session token manipulation and OAuth 2.0 flow abuse. Experience with both offensive testing and defensive program hardening.

Bison Parser with AST Construction | *C, Flex, Bison* **Fall 2025**

- Implemented a complete parser for a C-like language using Flex and Bison, constructing and traversing abstract syntax trees (ASTs) with support for unary expressions, increment/decrement operators, else-if chaining, array declarations, and for-loop variable declarations.
- Gained hands-on experience with formal grammar specification, shift-reduce conflict resolution, and memory-safe AST management. Foundational to compiler-level program analysis.

Distributed Hash Table (DHT) Implementation | *Python, Sockets, Threading* **May 2025**

- Implemented a Chord-like DHT with consistent hashing for decentralized key-value storage and fault-tolerant key lookup across dynamic peer-to-peer nodes.

Technical Skills

Languages: Python, C++, C, Go, MATLAB
Security Tools: GDB, BurpSuite, Wireshark
Analysis Techniques: Static binary analysis, dynamic instrumentation, exploit development, API/system call tracing
Developer Tools: VSCode, Docker, Git
Technologies: Linux, Socket programming, TCP/IP, Threading

Achievements and Awards

- Placed on the LUMS Dean's Honor List for Academic Year 2023–24 (Freshman Year)
- Placed on the LUMS Dean's Honor List for Academic Year 2024–25 (Sophomore Year)
- Placed on the LUMS Dean's Honor List for Academic Year 2025–26 (Junior Year)
- Placed on the LUMS Dean's Honor List for Academic Year 2026–27 (Senior Year)

Certificates

- Cybersecurity Professional Certificate – Google
- Introduction to Cybersecurity – Cisco
- Fundamentals of Deep Learning – Nvidia
- Advanced Deep Learning Techniques for Computer Vision – MathWorks
- Deep Learning for Object Detection – MathWorks
- Getting Started with Go on Google Cloud – Google
- Teaching Assistant Professional Development Course – LUMS